OpenDRIVE® 1.8.0, Static Road Network Description, 2023-11-22

9 Traffic lights

9.1 Introduction

A junction can contain a number of different traffic lights and corresponding stop lines as well as other signals that interact with the traffic participants and have to be harmonized. Modeling the relations between the different elements to let the simulation and traffic run smoothly can be challenging. Depending on the country, there can be different placements of the various elements within the junction.

In ASAM OpenDRIVE, the `<signal>` element can define relations as well as dependencies to other signals or objects through its attributes, various child elements, and their attributes. The following examples will show our recommendations for different situations.

§ 9.2 Rules and guidelines

- Traffic lights shall be defined at their physical location with the corresponding s-/t-coordinates of the road. This is not necessarily the road on which the traffic participant shall stop for the traffic light.
- A traffic light itself does not tell you where to stop, instead a stop line does. On the other hand, a stop line is only of interest if the traffic light is red (or off). Therefore, the stop line shall have a dependency to each of the corresponding traffic lights:
  `<dependency id="…" type="trafficLight" />`
- Signs that interact with traffic lights also may have a dependency, but not necessarily.
- The traffic light shall have a reference to the relevant stop line:
  `<reference id="…" elementType="signal" type="stopLine" />`
- Multiple traffic lights may share the same stop line, resulting in multiple references.

- The traffic light shall not get a `laneValidity` element. The `laneValidity` element of the referenced stop line shall be taken: `<validity fromLane="…" toLane="…" />`
- All traffic lights that have the same phases shall share a common controller, for example, green arrows.
- Traffic rules are not part of the ASAM OpenDRIVE definition itself but can have an impact on the relations of those elements.

9.3 Road with two traffic lights and one stop line

9.3.1 Example

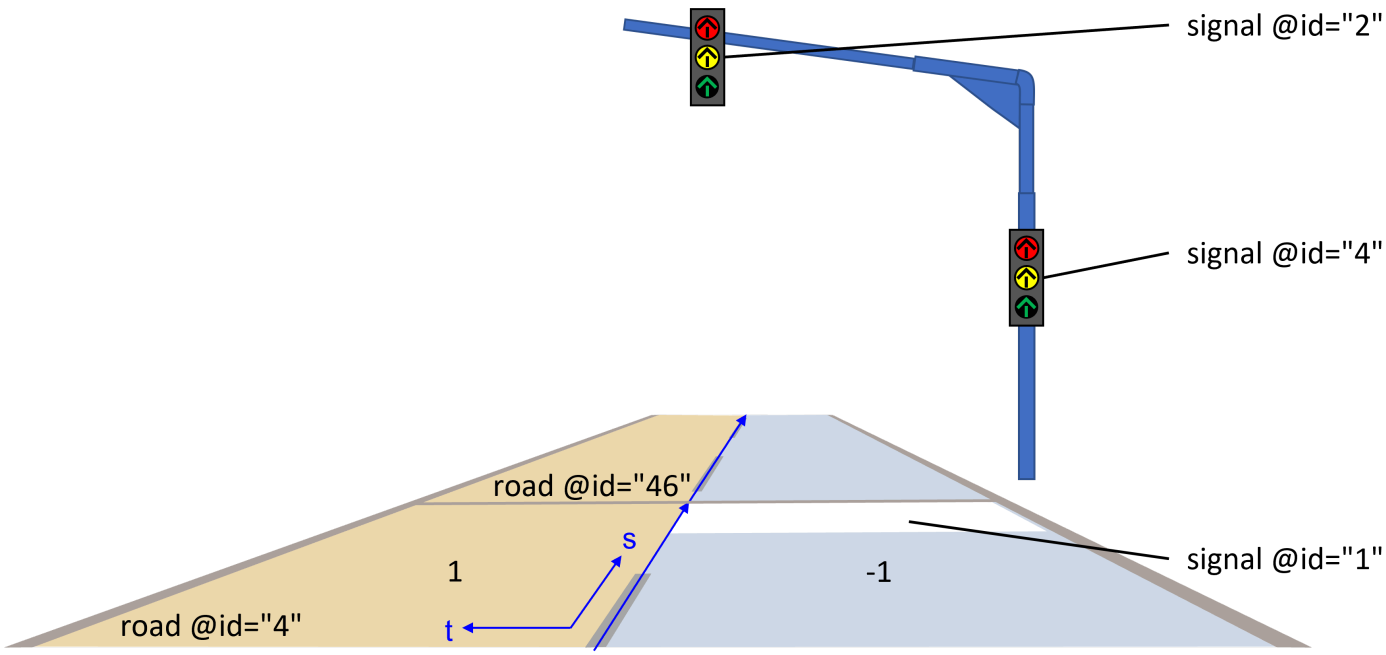

Figure 55. Roads with traffic lights and stop line

Figure 55 shows two roads with two traffic lights and one stop line:

- The road with @id="4" is a two-way road with a single lane with @type="driving" for each direction. On the road with @id="4", there is one stop line signal with @id="1" on the lane with @id="-1".

- The road with @id="46" is a two-way road with a single lane with @type="driving" for each direction. On the road with @id="46", there are two traffic lights, the signal with @id="2" is above the road and the signal with @id="4" is to the right on the sidewalk.

9.3.2 Challenges

- The stop line signal with @id="1" is valid for two traffic lights, the signals with @id="2" and @id="4".
- The stop line signal with @id="1" and the two traffic lights, the signals with @id="2" and @id="4", are not on the same lane or road.

9.3.3 Modeling approach

1. Set the validity for the signal with @id="1" on the road with @id="4" to the lane with @id="-1".

Code 37. `<validity>` element with @fromLane and @toLane attributes in the `<signal>` element with @id="1"

```
<road name="" length="6.8257399999999990e+01" id="4" junction="-1" rule="RHT">
    ...
   <signals>
        <signal s="6.8000000000000000e+01"
                t="-0.0000000000000000e+00"
                id="1"
                name="StopLine"
                type="Haltelinie"
                ...>
            <validity fromLane="-1" toLane="-1"/>
        </signal>
    </signals>
</road>
```

2. Set the dependency for the signal with @id="1" on the road with @id="4" to the signals with @id="2" and @id="4" on the road with @id="46".

Code 38. `<dependency>` elements in the `<signal>` element with @id="1" on the road with @id="4" pointing to the signals with @id="2" and @id="4" on the road with @id="46"

```xml
<road name="" length="6.8257399999999990e+01" id="4" junction="-1" rule="RHT">
    ...
   <signals>
        <signal s="6.8000000000000000e+01"
                t="-0.0000000000000000e+00"
                id="1"
                name="StopLine"
                type="Haltelinie"
                ...>
            <validity fromLane="-1" toLane="-1"/>
            <dependency id="2" type="trafficLight"/>
            <dependency id="4" type="trafficLight"/>
        </signal>
    </signals>
</road>
<road name="" length="3.9995576531488346e+01" id="46" junction="10" rule="RHT">
    ...
    <signals>
        <signal s="2.5000000000000000e+00"
                t="-1.8400000000000001e+00"
                id="2"
                name="TrafficLight"
                type="1000011"
                ...>
        </signal>
        <signal s="2.5000000000000000e+00"
                t="-4.0000000000000000e+00"
                id="4"
                name="TrafficLight"
                type="1000011"
                ...>
        </signal>
    </signals>
    ...
</road>
```

```
3
2
4
2
5
2
6
2
7
2
8
2
9
3
0
3
1
3
2
3
3
3
4
3
5
```

3. Set the reference for the signals with @id="2" and @id="4" on the road with @id="46" to the signal with @id="1" on the road with @id="4".

Code 39. `<reference>` elements in the `<signal>` elements with @id="2" and @id="4" on the road with @id="46" pointing to the signal with @id="1" on the road with @id="4"

```
<road name="" length="6.8257399999999990e+01" id="4" junction="-1" rule="RHT">
    ...
   <signals>
        <signal s="6.8000000000000000e+01"
                t="-0.0000000000000000e+00"
                id="1"
                name="StopLine"
                type="Haltelinie"
                ...>
            <validity fromLane="-1" toLane="-1"/>
            <dependency id="2" type="trafficLight"/>
            <dependency id="4" type="trafficLight"/>
        </signal>
    </signals>
</road>
<road name="" length="3.9995576531488346e+01" id="46" junction="10" rule="RHT">
    ...
    <signals>
        <signal s="2.5000000000000000e+00"
                t="-1.8400000000000001e+00"
                id="2"
                name="TrafficLight"
                type="1000011"
                ...>
            <reference id="1" elementType="signal" type="stopLine"/>
        </signal>
        <signal s="2.5000000000000000e+00"
                t="-4.0000000000000000e+00"
                id="4"
                name="TrafficLight"
                type="1000011"
                ...>
            <reference id="1" elementType="signal" type="stopLine"/>
        </signal>
    </signals>

```

```
    ...
</road>
```

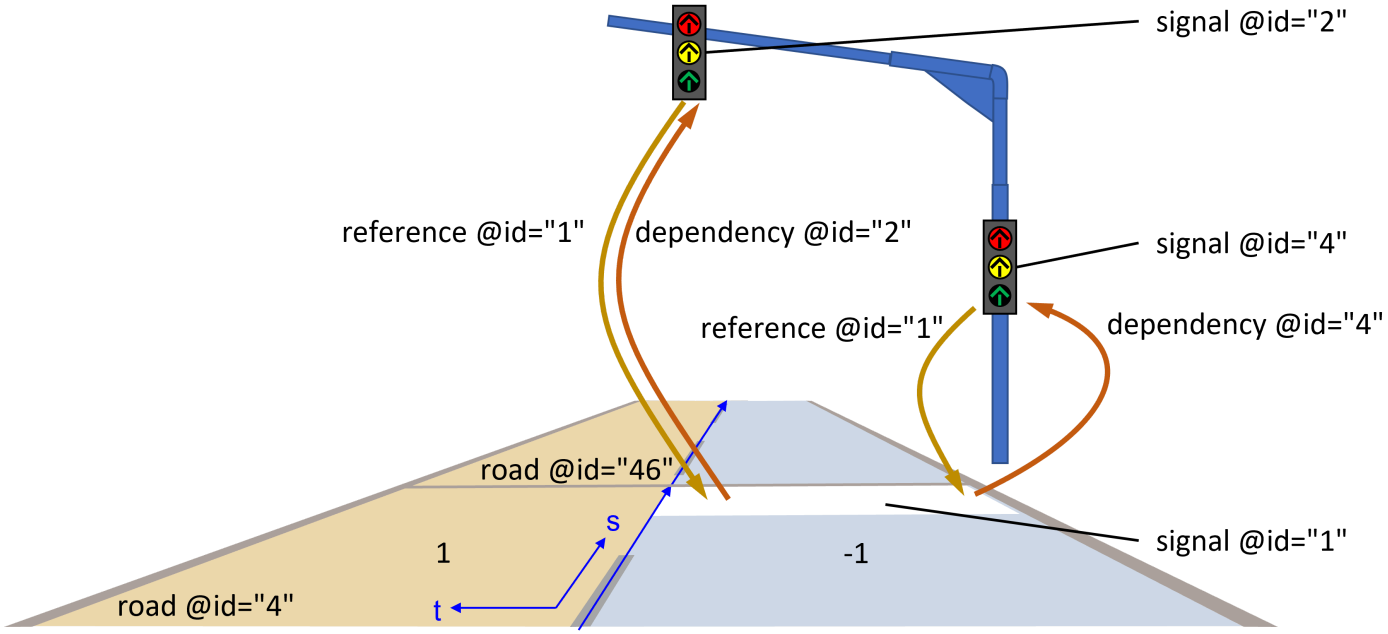

Figure 56. References and dependencies between traffic lights and stop line

Figure 56 shows the `<dependency>` elements of the traffic lights pointing to the stop line and the `<reference>` elements of the stop line pointing to the traffic lights.

XML example

Code 40. Solution in ASAM OpenDRIVE 1.8

```
<road name=""
      length="6.8257399999999990e+01"
      id="4"
      junction="-1"
      rule="RHT">
    <signals>
        <signal s="6.8000000000000000e+01"
                t="-0.0000000000000000e+00"
                id="1"
                name="StopLine"
                dynamic="no"
                orientation="+"
                zOffset="0.0000000000000000e+00"
                type="Haltelinie"
                country="AT"
                countryRevision="2013"
                subtype="-1"
                hOffset="0.0000000000000000e+00"
                pitch="0.0000000000000000e+00"
                roll="0.0000000000000000e+00"
                height="0.03"
                width="3.40">
            <validity fromLane="-1" toLane="-1"/>
            <dependency id="2" type="trafficLight"/>
            <dependency id="4" type="trafficLight"/>
        </signal>
    </signals>
</road>
<road name=""
      length="3.9995576531488346e+01"
      id="46"
      junction="10"
      rule="RHT">
    ...
    <signals>
        <signal s="2.5000000000000000e+00"
```

```
                t="-1.8400000000000001e+00"
                id="2"
                name="TrafficLight"
                dynamic="yes"
                orientation="+"
                zOffset="4.4200000166893005e+00"
                type="1000011"
                country="OpenDRIVE"
                countryRevision="2013"
                subtype="-1"
                hOffset="6.2831853071795862e+00"
                pitch="0.0000000000000000e+00"
                roll="0.0000000000000000e+00"
                height="1.15"
                width="0.41">
            <reference id="1" elementType="signal" type="stopLine"/>
        </signal>
        <signal s="2.5000000000000000e+00"
                t="-4.0000000000000000e+00"
                id="4"
                name="TrafficLight"
                dynamic="yes"
                orientation="+"
                zOffset="2.4200000166893005e+00"
                type="1000011"
                country="OpenDRIVE"
                countryRevision="2013"
                subtype="-1"
                hOffset="6.2831853071795862e+00"
                pitch="0.0000000000000000e+00"
                roll="0.0000000000000000e+00"
                height="1.15"
                width="0.41">
            <reference id="1" elementType="signal" type="stopLine"/>
        </signal>
    </signals>

```

```
    ...
</road>
```

```
0
6
1
6
2
6
3
6
4
6
5
6
6
6
7
6
8
6
9
7
0
7
1
7
2
7
3
7
4
```

- UC_X_Junction.xodr

9.4 Junction with several traffic lights for multiple stop lines

9.4.1 Example

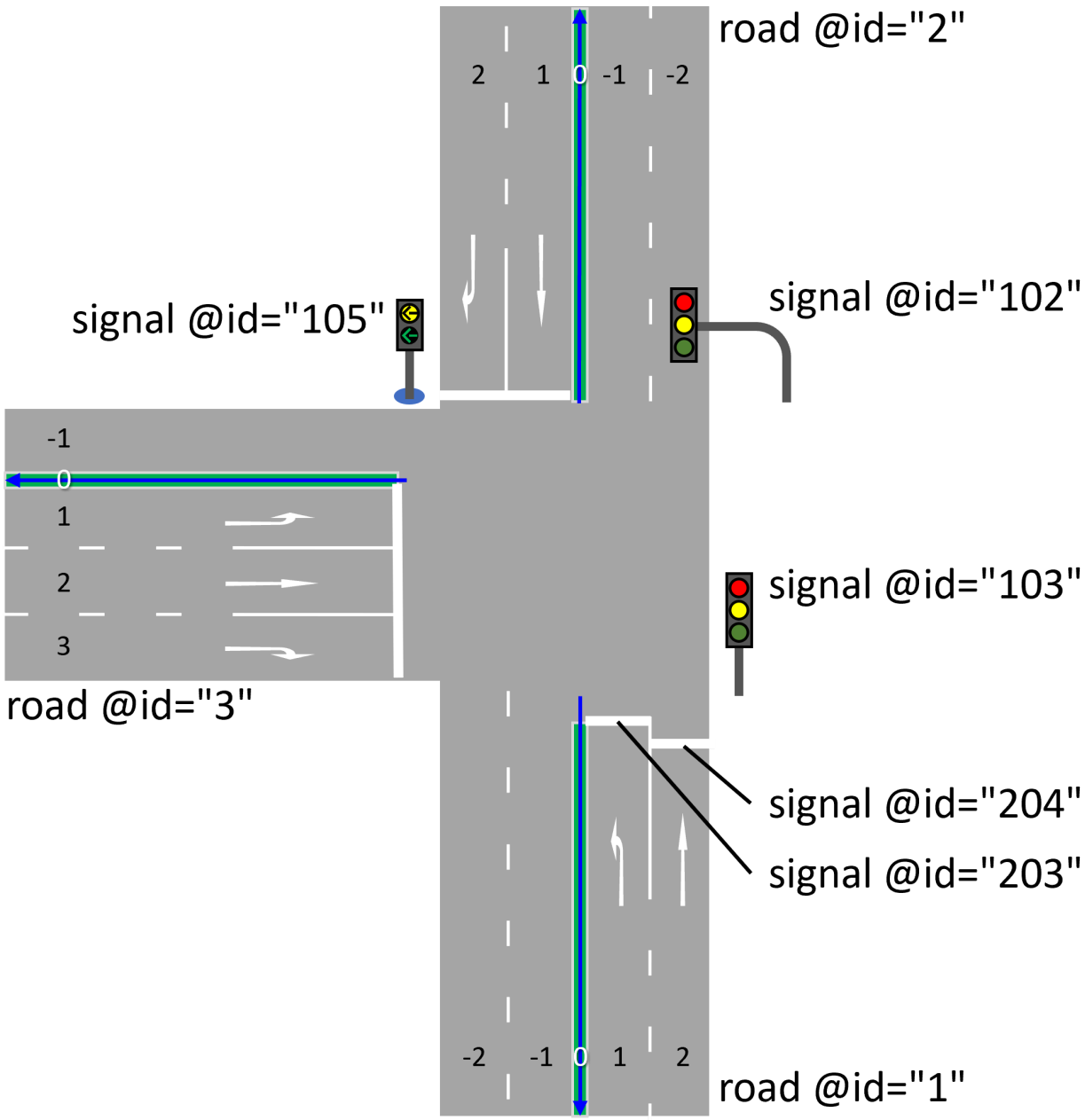

Figure 57. T-junction with three traffic lights and two stop lines

Figure 57 shows a T-junction with several roads, lanes, traffic lights, and stop lines:

- The road with @id="1" is a two-way road with two lanes with @type="driving" for each direction. On the road with @id="1", there are two separate stop lines, the signal with @id="203" for turning left on the lane with @id="1" and the signal with @id="204" for driving straight ahead on the lane with @id="2". Next to the road with @id="1", there is a traffic light ,the signal with @id="103".
- The road with @id="2" is a two-way road with two lanes with @type="driving" for each direction. Next to the road with @id="2", there is a traffic light, the signal with @id="102".
- The road with @id="3" is a two-way road with three lanes with @type="driving" on the right and one lane with @type="driving" on the left. Next to the road with @id="3", there is a traffic light, the signal with @id="105".

9.4.2 Challenges

- The stop line signal with @id="203" for turning left is valid for all three traffic light signals with @id="102", @id="103", and @id="105".
- The stop line signal with @id="204" for driving straight ahead is valid for two of the traffic lights, the signals with @id="102" and @id="103".
- Only one of the traffic lights, ths signal with @id="103", and the two stop lines, the signals with @id="203" and @id="204", are on the same road. The other two traffic lights, the signals with @id="102" and @id="105", are behind the junction.

9.4.3 Modeling approach

1. Set the validity for the stop line signal with @id="203" on the road with @id="1" to the lane with @id="1".

Code 41. `<validity>` element with @fromLane and @toLane attributes set to `1` in the `<signal>` element with @id="203"

```xml
<road name="" length="5.0000000000000000e+01" id="1" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="3.0000000000000000e+00"
                t="0.0000000000000000e+00"
                id="203"
                name="StopLineLeft"
                dynamic="no"
                orientation="-"
                zOffset="0.0000000000000000e+00"
                type="294"
                country="OpenDRIVE"
                countryRevision="2013"
                subtype="-1"
                value="0.0000000000000000e+00"
                hOffset="0.0000000000000000e+00"
                pitch="0.0000000000000000e+00"
                roll="0.0000000000000000e+00"
                height="0.03"
                width="2.80">
            <validity fromLane="1" toLane="1" />
        </signal>
    </signals>
</road>
```

```
3
2
4
```

2. Set the dependency for the stop line signal with @id="203" on the road with @id="1" to the traffic light signals with @id="102", @id="103" and @id="105".

Code 42. `<dependency>` elements in the `<signal>` element with @id="203" on the road with @id="1" pointing to the `<signal>` element with @id="102" on the road with @id="2", the `<signal>` element with @id="103" on the road with @id="1", and the `<signal>` element with @id="105" on the road with @id="3".

```xml
<road name="" length="5.0000000000000000e+01" id="1" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="3.0000000000000000e+00"
                t="0.0000000000000000e+00"
                id="203"
                name="StopLineLeft"
                type="294"
                ...>
            <validity fromLane="1" toLane="1"/>
            <dependency id="102" type="trafficLight" />
            <dependency id="103" type="trafficLight" />
            <dependency id="105" type="trafficLight" />
        </signal>
        ...
        <signal s="0.0000000000000000e+00"
                t="-1.1000000000000000e+01"
                id="103"
                name="TrafficLight_red_yellow_green"
                type="1000001"
                ...>
        </signal>
    </signals>
</road>
<road name="" length="5.0000000000000000e+01" id="2" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="0.0000000000000000e+00"
                t="-0.0000000000000000e+00"
                id="102"
                name="TrafficLight_red_yellow_green"
                type="1000001"
                ...>
        </signal>
    </signals>
</road>
```

```
<road name="" length="5.0000000000000000e+01" id="3" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="0.0000000000000000e+00"
                t="5.0000000000000000e+00"
                id="105"
                name="TrafficLight_yellow_green"
                type="1000009"
                ...>
        </signal>
    </signals>
</road>
```

```
4
2
4
3
4
4
4
5
4
6
4
7
4
8
```

3. Set the validity for the stop line signal with @id="204" on the road with @id="1" to the lane with @id="2".

Code 43. `<validity>` element with @fromLane and @toLane attributes set to `2` in the `<signal>` element with @id="204"

```xml
<road name="" length="5.0000000000000000e+01" id="1" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="3.0000000000000000e+00"
                t="0.0000000000000000e+00"
                id="203"
                name="StopLineLeft"
                type="294"
                ...>
            <validity fromLane="1" toLane="1"/>
            <dependency id="102" type="trafficLight" />
            <dependency id="103" type="trafficLight" />
            <dependency id="105" type="trafficLight" />
        </signal>
        <signal s="4.0000000000000000e+00"
                t="0.0000000000000000e+00"
                id="204"
                name="StopLineStraight"
                type="294"
                ...>
            <validity fromLane="2" toLane="2" />
        </signal>
        <signal s="0.0000000000000000e+00"
                t="-1.1000000000000000e+01"
                id="103"
                name="TrafficLight_red_yellow_green"
                type="1000001"
                ...>
        </signal>
    </signals>
</road>
```

```
3
2
4
2
5
2
6
2
7
2
8
2
9
3
0
3
1
```

4. Set the dependency for the stop line signal with @id="204" on the road with @id="1" to the traffic light signals with @id="102" and @id="103".

Code 44. `<dependency>` elements in the `<signal>` element with @id="204" on the road with @id="1" pointing to the `<signal>` element with @id="102" on the road with @id="2", and the `<signal>` element with @id="103" on the road with @id="1"

```xml
<road name="" length="5.0000000000000000e+01" id="1" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="3.0000000000000000e+00"
                t="0.0000000000000000e+00"
                id="203"
                name="StopLineLeft"
                type="294"
                ...>
            <validity fromLane="1" toLane="1"/>
            <dependency id="102" type="trafficLight" />
            <dependency id="103" type="trafficLight" />
            <dependency id="105" type="trafficLight" />
        </signal>
        <signal s="4.0000000000000000e+00"
                t="0.0000000000000000e+00"
                id="204"
                name="StopLineStraight"
                type="294"
                ...>
            <validity fromLane="2" toLane="2" />
            <dependency id="102" type="trafficLight" />
            <dependency id="103" type="trafficLight" />
        </signal>
        ...
        <signal s="0.0000000000000000e+00"
                t="-1.1000000000000000e+01"
                id="103"
                name="TrafficLight_red_yellow_green"
                type="1000001"
                ...>
        </signal>
    </signals>
</road>
<road name="" length="5.0000000000000000e+01" id="2" junction="-1" rule="RHT">
    ...
```

```
    <signals>
        <signal s="0.0000000000000000e+00"
                t="-0.0000000000000000e+00"
                id="102"
                name="TrafficLight_red_yellow_green"
                type="1000001"
                ...>
        </signal>
    </signals>
</road>
<road name="" length="5.0000000000000000e+01" id="3" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="0.0000000000000000e+00"
                t="5.0000000000000000e+00"
                id="105"
                name="TrafficLight_yellow_green"
                type="1000009"
                ...>
        </signal>
    </signals>
</road>
```

```
4
2
4
3
4
4
4
5
4
6
4
7
4
8
4
9
5
0
5
1
5
2
5
3
5
4
5
5
5
6
5
7
5
8
```

5. Set the reference for the traffic light signal with @id="103" on the road with @id="1" to the stop line signals with @id="203" and @id="204".

Code 45. `<reference>` elements in the `<signal>` element with @id="103" on the road with @id="1" pointing to the `<signal>` elements with @id="203" and @id="204" on the road with @id="1"

```xml
<road name="" length="5.0000000000000000e+01" id="1" junction="-1" rule="RHT">
    ...
   <signals>
        <signal s="3.0000000000000000e+00"
                t="0.0000000000000000e+00"
                id="203"
                name="StopLineLeft"
                type="294"
                ...>
            <validity fromLane="1" toLane="1"/>
            <dependency id="102" type="trafficLight" />
            <dependency id="103" type="trafficLight" />
            <dependency id="105" type="trafficLight" />
        </signal>
        <signal s="4.0000000000000000e+00"
                t="0.0000000000000000e+00"
                id="204"
                name="StopLineStraight"
                type="294"
                ...>
            <validity fromLane="2" toLane="2" />
            <dependency id="102" type="trafficLight" />
            <dependency id="103" type="trafficLight" />
        </signal>
        ...
        <signal s="0.0000000000000000e+00"
                t="-1.1000000000000000e+01"
                id="103"
                name="TrafficLight_red_yellow_green"
                type="1000001"
                ...>
            <reference id="203" elementType="signal" type="stopLine" />
            <reference id="204" elementType="signal" type="stopLine" />
        </signal>
    </signals>
</road>
```

```
<road name="" length="5.0000000000000000e+01" id="2" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="0.0000000000000000e+00"
                t="-0.0000000000000000e+00"
                id="102"
                name="TrafficLight_red_yellow_green"
                type="1000001"
                ...>
        </signal>
    </signals>
</road>
<road name="" length="5.0000000000000000e+01" id="3" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="0.0000000000000000e+00"
                t="5.0000000000000000e+00"
                id="105"
                name="TrafficLight_yellow_green"
                type="1000009"
                ...>
        </signal>
    </signals>
</road>
```

```
4
2
4
3
4
4
4
5
4
6
4
7
4
8
4
9
5
0
5
1
5
2
5
3
5
4
5
5
5
6
5
7
5
8
5
9
```

```
6
0
```

6. Set the reference for the traffic light signal with @id="102" on the road with @id="2" to the stop line signals with @id="203" and @id="204".

Code 46. `<reference>` elements in the `<signal>` element with @id="102" on the road with @id="2" pointing to the `<signal>` elements with @id="203" and @id="204" on the road with @id="1"

```xml
<road name="" length="5.0000000000000000e+01" id="1" junction="-1" rule="RHT">
    ...
   <signals>
        <signal s="3.0000000000000000e+00"
                t="0.0000000000000000e+00"
                id="203"
                name="StopLineLeft"
                type="294"
                ...>
            <validity fromLane="1" toLane="1"/>
            <dependency id="102" type="trafficLight" />
            <dependency id="103" type="trafficLight" />
            <dependency id="105" type="trafficLight" />
        </signal>
        <signal s="4.0000000000000000e+00"
                t="0.0000000000000000e+00"
                id="204"
                name="StopLineStraight"
                type="294"
                ...>
            <validity fromLane="2" toLane="2" />
            <dependency id="102" type="trafficLight" />
            <dependency id="103" type="trafficLight" />
        </signal>
        ...
        <signal s="0.0000000000000000e+00"
                t="-1.1000000000000000e+01"
                id="103"
                name="TrafficLight_red_yellow_green"
                type="1000001"
                ...>
            <reference id="203" elementType="signal" type="stopLine" />
            <reference id="204" elementType="signal" type="stopLine" />
        </signal>
    </signals>
</road>
```

```
<road name="" length="5.0000000000000000e+01" id="2" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="0.0000000000000000e+00"
                t="-0.0000000000000000e+00"
                id="102"
                name="TrafficLight_red_yellow_green"
                type="1000001"
                ...>
            <reference id="203" elementType="signal" type="stopLine" />
            <reference id="204" elementType="signal" type="stopLine" />
        </signal>
    </signals>
</road>
<road name="" length="5.0000000000000000e+01" id="3" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="0.0000000000000000e+00"
                t="5.0000000000000000e+00"
                id="105"
                name="TrafficLight_yellow_green"
                type="1000009"
                ...>
        </signal>
    </signals>
</road>
```

```
4
2
4
3
4
4
4
5
4
6
4
7
4
8
4
9
5
0
5
1
5
2
5
3
5
4
5
5
5
6
5
7
5
8
5
9
6
```

```
0
6
1
6
2
```

7. Set the reference for the traffic light signal with @id="105" on the road with @id="3" to the stop line signal with @id="203".

Code 47. `<reference>` element in the `<signal>` element with @id="105" on the road with @id="3" pointing to the `<signal>` element with @id="203" on the road with @id="1"

```xml
<road name="" length="5.0000000000000000e+01" id="1" junction="-1" rule="RHT">
    ...
   <signals>
        <signal s="3.0000000000000000e+00"
                t="0.0000000000000000e+00"
                id="203"
                name="StopLineLeft"
                type="294"
                ...>
            <validity fromLane="1" toLane="1"/>
            <dependency id="102" type="trafficLight" />
            <dependency id="103" type="trafficLight" />
            <dependency id="105" type="trafficLight" />
        </signal>
        <signal s="4.0000000000000000e+00"
                t="0.0000000000000000e+00"
                id="204"
                name="StopLineStraight"
                type="294"
                ...>
            <validity fromLane="2" toLane="2" />
            <dependency id="102" type="trafficLight" />
            <dependency id="103" type="trafficLight" />
        </signal>
        ...
        <signal s="0.0000000000000000e+00"
                t="-1.1000000000000000e+01"
                id="103"
                name="TrafficLight_red_yellow_green"
                type="1000001"
                ...>
            <reference id="203" elementType="signal" type="stopLine" />
            <reference id="204" elementType="signal" type="stopLine" />
        </signal>
    </signals>
</road>
```

```
<road name="" length="5.0000000000000000e+01" id="2" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="0.0000000000000000e+00"
                t="-0.0000000000000000e+00"
                id="102"
                name="TrafficLight_red_yellow_green"
                type="1000001"
                ...>
            <reference id="203" elementType="signal" type="stopLine" />
            <reference id="204" elementType="signal" type="stopLine" />
        </signal>
    </signals>
</road>
<road name="" length="5.0000000000000000e+01" id="3" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="0.0000000000000000e+00"
                t="5.0000000000000000e+00"
                id="105"
                name="TrafficLight_yellow_green"
                type="1000009"
                ...>
            <reference id="203" elementType="signal" type="stopLine" />
        </signal>
    </signals>
</road>
```

```
4
2
4
3
4
4
4
5
4
6
4
7
4
8
4
9
5
0
5
1
5
2
5
3
5
4
5
5
5
6
5
7
5
8
5
9
6
```

```
0
6
1
6
2
6
3
```

XML example

Code 48. Solution in ASAM OpenDRIVE 1.8.0

```
<road name="" length="5.0000000000000000e+01" id="1" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="3.0000000000000000e+00"
                t="0.0000000000000000e+00"
                id="203"
                name="StopLineLeft"
                dynamic="no"
                orientation="-"
                zOffset="0.0000000000000000e+00"
                type="294"
                country="OpenDRIVE"
                countryRevision="2013"
                subtype="-1"
                value="0.0000000000000000e+00"
                hOffset="0.0000000000000000e+00"
                pitch="0.0000000000000000e+00"
                roll="0.0000000000000000e+00"
                height="0.03"
                width="2.80">
            <validity fromLane="1" toLane="1" />
            <dependency id="102" type="trafficLight" />
            <dependency id="103" type="trafficLight" />
            <dependency id="105" type="trafficLight" />
        </signal>
        <signal s="4.0000000000000000e+00"
                t="0.0000000000000000e+00"
                id="204"
                name="StopLineStraight"
                dynamic="no"
                orientation="-"
                zOffset="0.0000000000000000e+00"
                type="294"
                country="OpenDRIVE"
                countryRevision="2013"
                subtype="-1"
```

```
                value="0.0000000000000000e+00"
                hOffset="0.0000000000000000e+00"
                pitch="0.0000000000000000e+00"
                roll="0.0000000000000000e+00"
                height="0.03"
                width="2.80">
            <validity fromLane="2" toLane="2" />
            <dependency id="102" type="trafficLight" />
            <dependency id="103" type="trafficLight" />
        </signal>
        <signal s="0.0000000000000000e+00"
                t="-1.1000000000000000e+01"
                id="103"
                name="TrafficLight_red_yellow_green"
                dynamic="yes"
                orientation="-"
                zOffset="2.4200000166893005e+00"
                type="1000001"
                country="OpenDRIVE"
                countryRevision="2013"
                subtype="-1"
                value="0.0000000000000000e+00"
                hOffset="0.0000000000000000e+00"
                pitch="0.0000000000000000e+00"
                roll="0.0000000000000000e+00"
                height="1.15"
                width="0.38">
            <reference id="203" elementType="signal" type="stopLine" />
            <reference id="204" elementType="signal" type="stopLine" />
        </signal>
    </signals>
</road>
<road name="" length="5.0000000000000000e+01" id="2" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="0.0000000000000000e+00"
                t="-0.0000000000000000e+00"
```

```
                id="102"
                name="TrafficLight_red_yellow_green"
                dynamic="yes"
                orientation="-"
                zOffset="2.4200000166893005e+00"
                type="1000001"
                country="OpenDRIVE"
                countryRevision="2013"
                subtype="-1"
                value="0.0000000000000000e+00"
                hOffset="0.0000000000000000e+00"
                pitch="0.0000000000000000e+00"
                roll="0.0000000000000000e+00"
                height="1.15"
                width="0.38">
            <reference id="203" elementType="signal" type="stopLine" />
            <reference id="204" elementType="signal" type="stopLine" />
        </signal>
    </signals>
</road>
<road name="" length="5.0000000000000000e+01" id="3" junction="-1" rule="RHT">
    ...
    <signals>
        <signal s="0.0000000000000000e+00"
                t="5.0000000000000000e+00"
                id="105"
                name="TrafficLight_yellow_green"
                dynamic="yes"
                orientation="-"
                zOffset="2.4200000166893005e+00"
                type="1000009"
                country="OpenDRIVE"
                countryRevision="2013"
                subtype="20"
                value="0.0000000000000000e+00"
                hOffset="0.0000000000000000e+00"
                pitch="0.0000000000000000e+00"
```

```
                roll="0.0000000000000000e+00"
                height="1.15"
                width="0.38">
            <reference id="203" elementType="signal" type="stopLine" />
        </signal>
    </signals>
</road>
```

```
7
9
8
0
8
1
8
2
8
3
8
4
8
5
8
6
8
7
8
8
8
9
9
0
9
1
9
2
9
3
9
4
9
5
9
6
9
```

```
7
9
8
9
9
1
0
0
1
0
1
1
0
2
1
0
3
1
0
4
1
0
5
1
0
6
1
0
7
1
0
8
1
0
9
1
1
```

```
0
1
1
1
1
1
2
1
1
3
1
1
4
1
1
5
1
1
6
1
1
7
```

9.5 Related topics

- ASAM OpenDRIVE Specification, Introduction to signals